

WHY DO WE NEED EXPOSOMICS ?

- 70 to 90% of causes of non chronic diseases are due to non-genetic factors
- Over 30 % of cancers are attributable to modifiable exposures
- Exposure measurements remain indirect and incomplete
- Cancer genomics identifies mutations- but not their environmental origin
- The genome tells us “what mutated” – not “what caused it.”

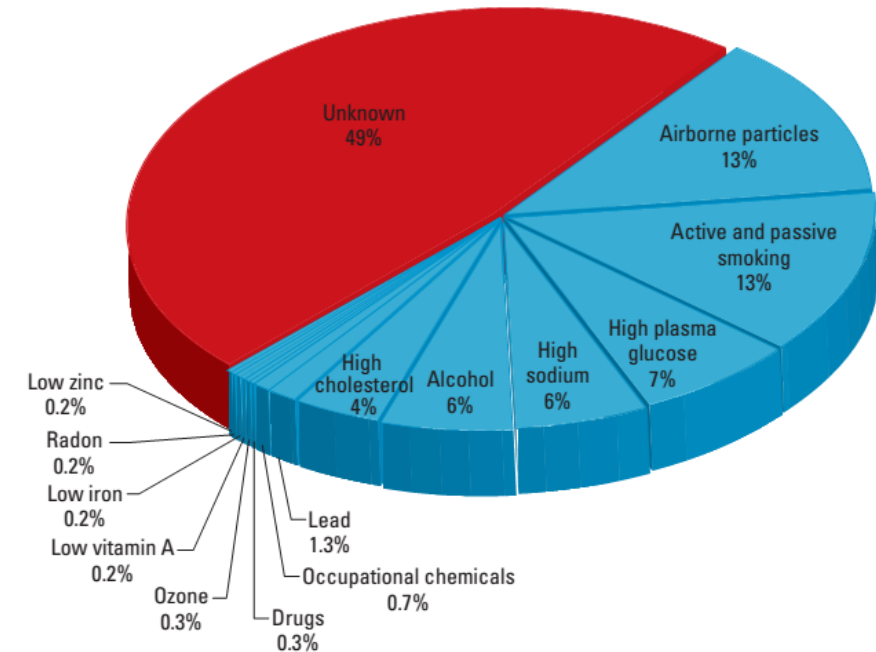
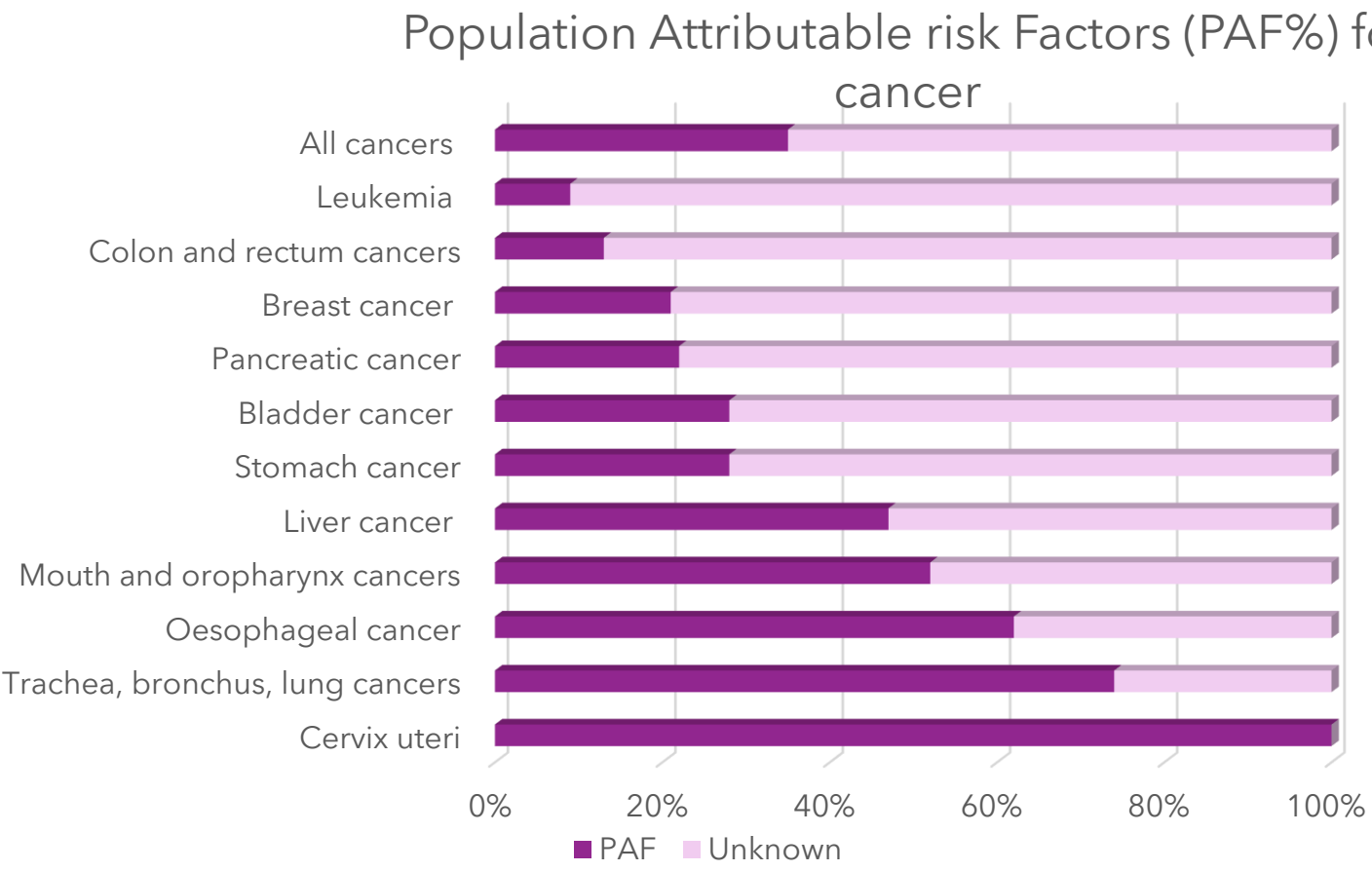


Figure 1. Risk factors for exposures that contribute to chronic-disease mortality. The chart was compiled from World Health Organization estimates of exposures affecting 50 million global deaths in 2010 (Lim et al. 2012). (Because some risk factors may be correlated, the indicated percentages are approximate.)

THE MISSING LAYER BETWEEN EXPOSURE AND MUTATION?



Find out more about the 12 recommendations:

- TOBACCO
- SECOND-HAND SMOKE
- HEALTHY BODY WEIGHT
- PHYSICAL ACTIVITY
- DIET
- ALCOHOL
- SUN/UV EXPOSURE
- POLLUTANTS
- RADIATION
- BREASTFEEDING
- HORMONAL THERAPY
- VACCINATION AND INFECTIONS
- SCREENING

Danaei et al. 2005, Causes of cancer in the world. The Lancet 366 (9499): 1784-1793
<https://cancer-code-europe.iarc.fr/index.php/en/ecac-12-ways>

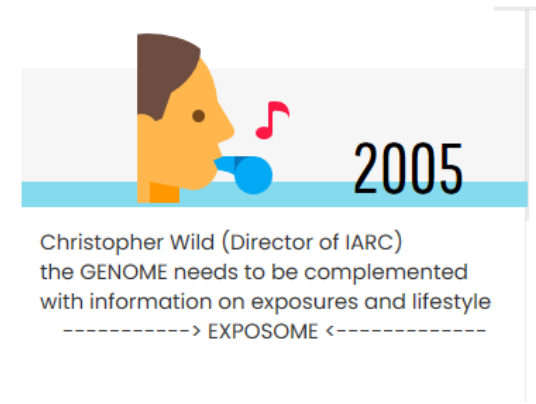
FROM GENOME TO EXPOSOME

Wild et al. 2005 "Complementing the Genome with an Exposome"

The concept of the exposome refers to the totality of environmental exposures **from conception onwards**, and includes both external and internal components.

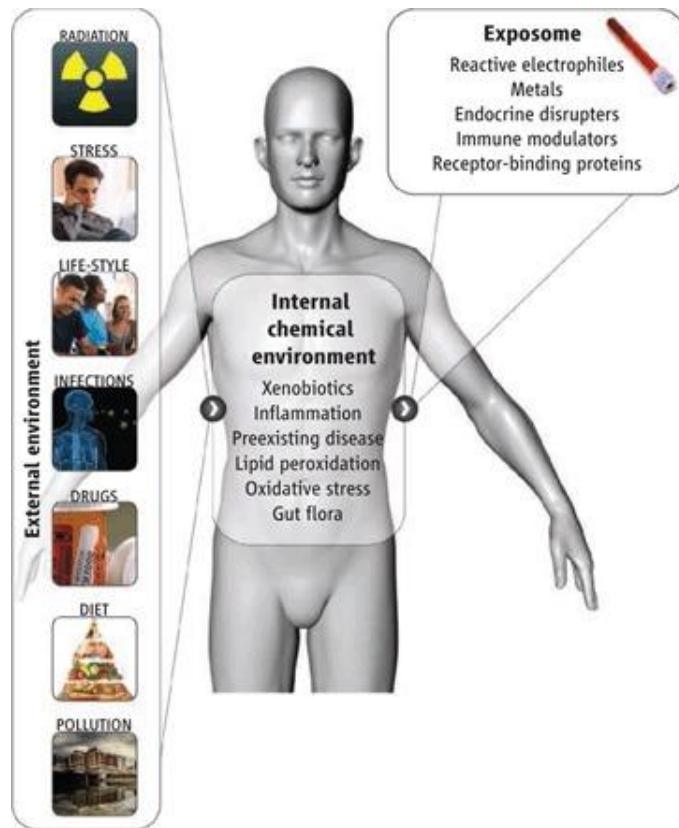


The history of the Exposome



FROM GENOME TO EXPOSOME

Rappaport and Smith 2010



The history of the Exposome



2005

Christopher Wild (Director of IARC)
the GENOME needs to be complemented
with information on exposures and lifestyle
-----> EXPOSOME <-----

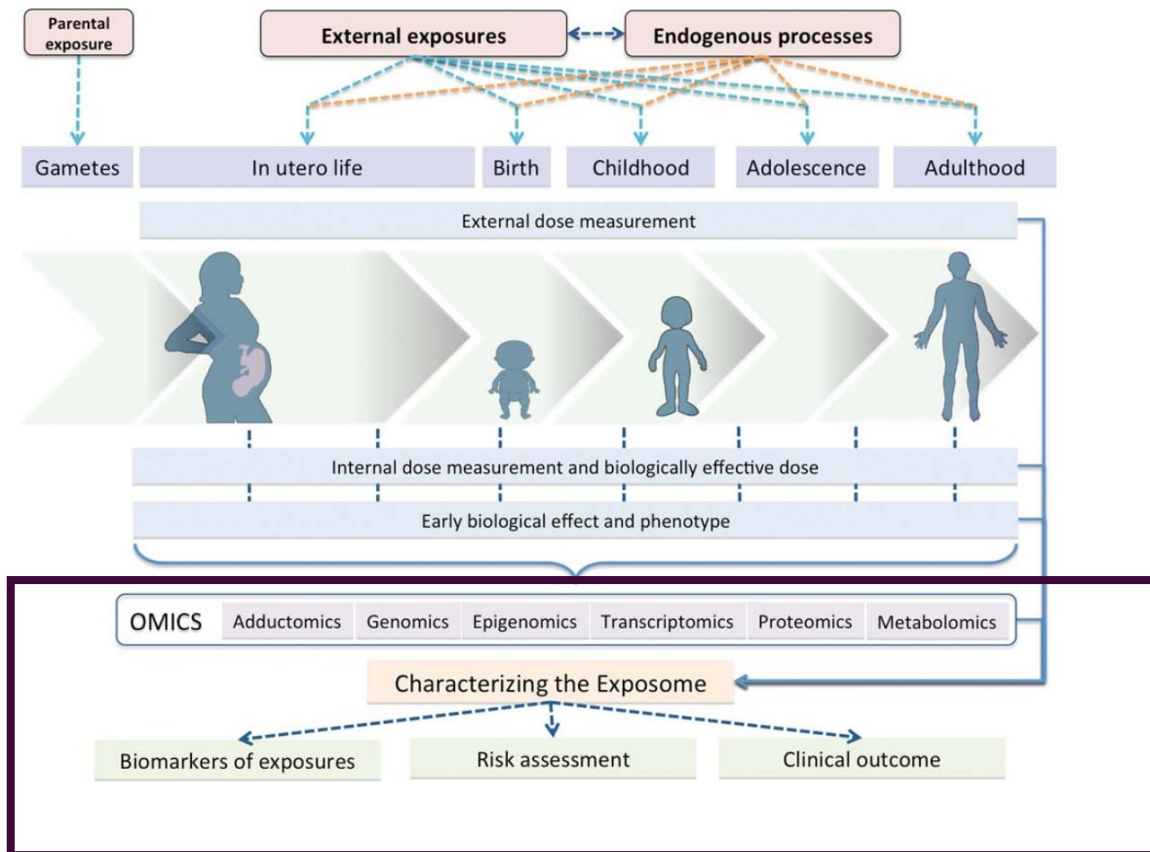


2010-2013

Reflection on the definition of the
Exposome and on the methods to
measure it

FROM GENOME TO EXPOSOME

Wild et al. 2013: Measuring the Exposome



The history of the Exposome



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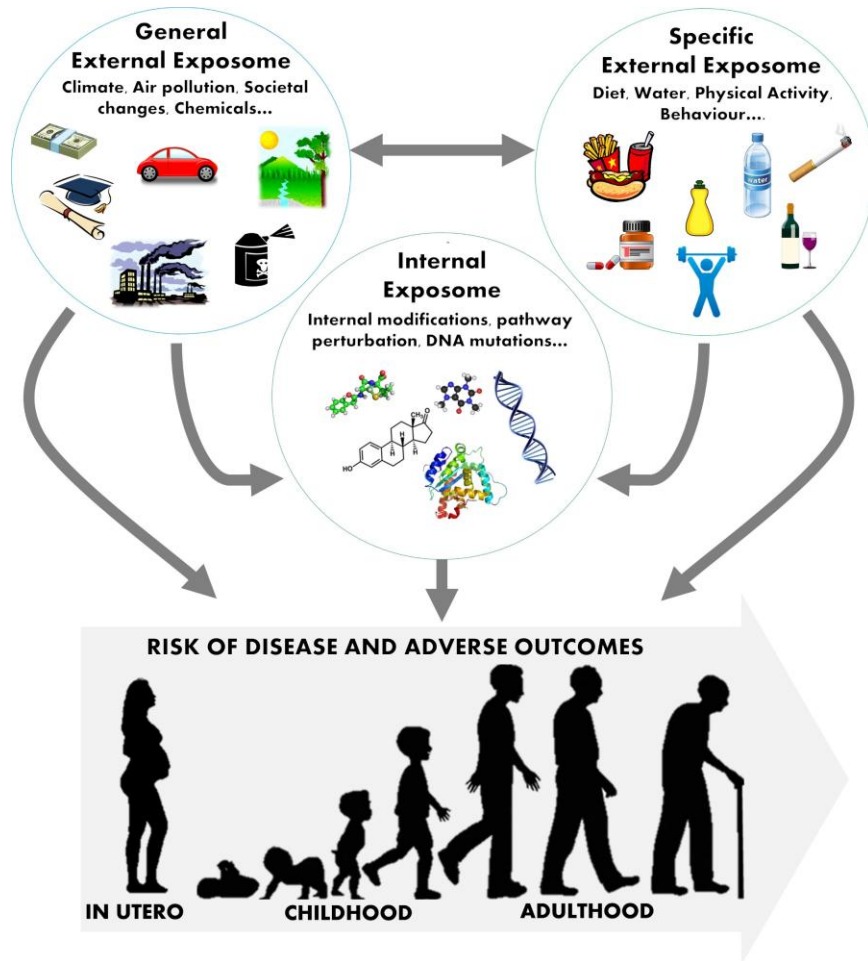
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THE EXPOSOME

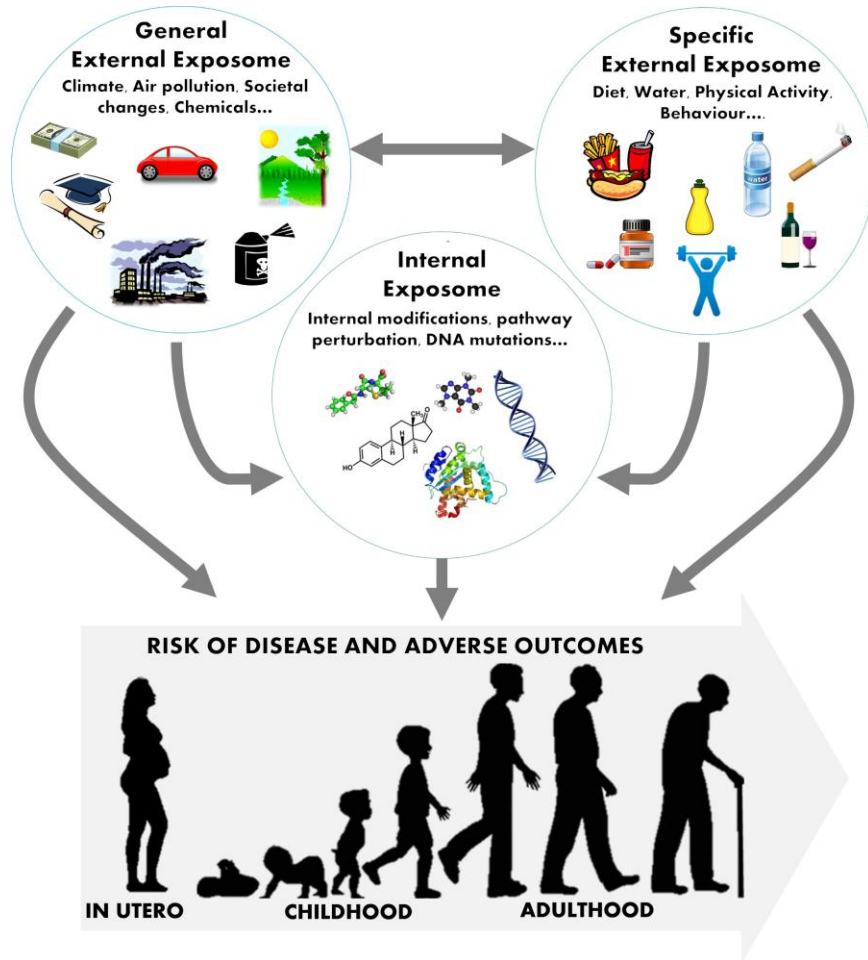


*"The cumulative measure of environmental influences and associated biological responses throughout the lifespan, including exposures from the environment, diet, behaviours and endogenous processes"**

From Dagnino S. and Macherone A. Unravelling the Exposome, Springer 2019.

*Miller and Jones (2014). The Nature of Nurture, refining the definition of the Exposome

THE EXPOSOME



The Exposome approach allows to

- ☐ Agnostic/systematic analysis of environment
- ☐ Improvement of exposure assessment
- ☐ Assessment of joint effects
- ☐ Improvement of mechanistic understanding
- ☐ Identification of common roots of diseases

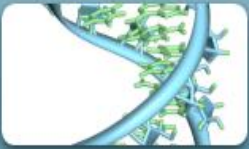
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THE INTERNAL EXPOSOME AND OMICS



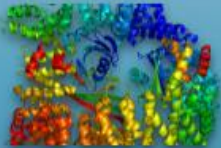
Epigenomic

- Measuring DNA methylation



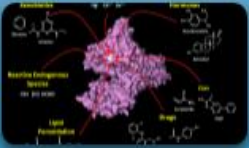
Transcriptomics

- Measuring mRNA



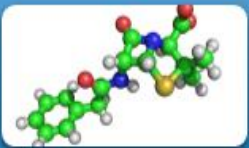
Proteomics

- Measuring proteins



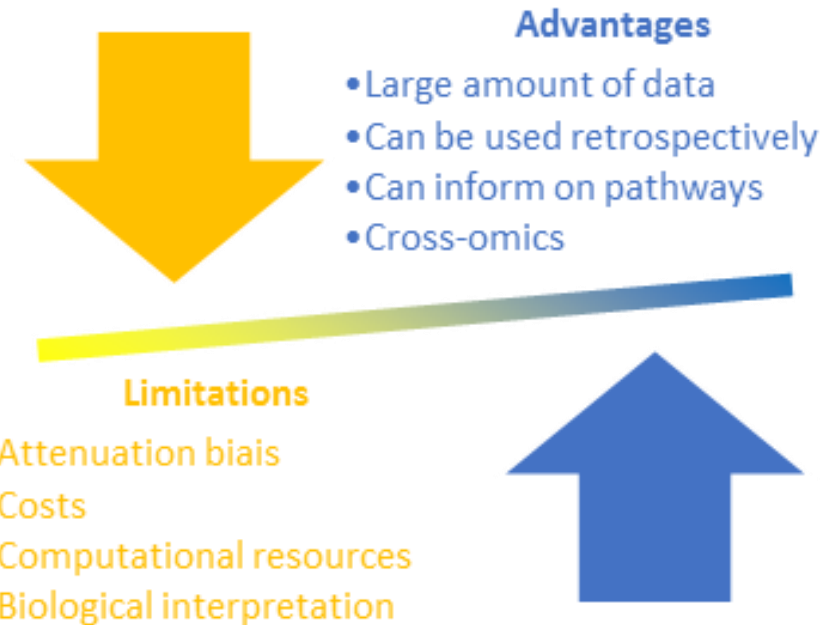
Adductomics

- Measuring adducts of proteins and DNA

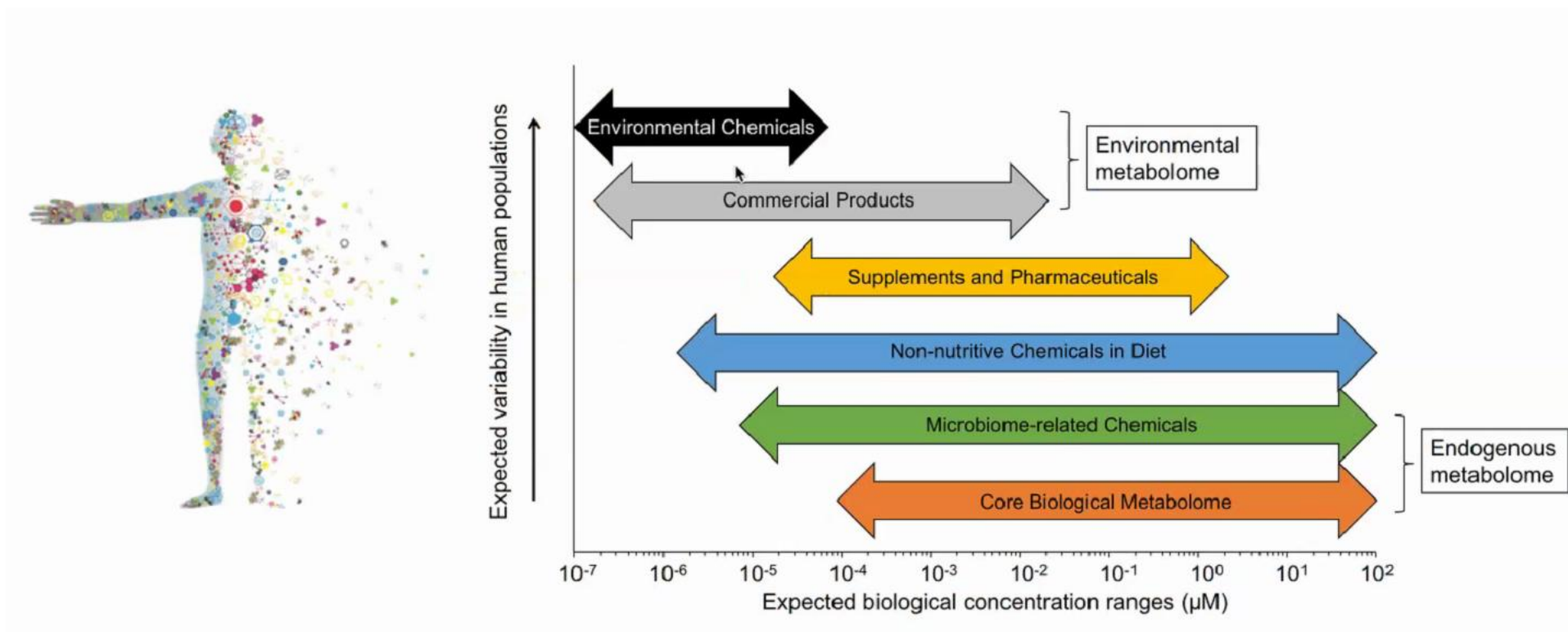


Metabolomics

- Measuring small molecules



METABOLOME FOR THE EXPOSOME



- Metabolomics allows the simultaneous measure of endogenous and exogenous metabolome.
- Essential and promising Omics for the Exposome

ADDUCTOMICS: THE NON-CANONICAL ENTRY POINT

Reactive Electrophiles: The Chemical Drivers of Mutation

Electrophiles are highly reactive species that have been long suspected of causing cancer and other diseases because of their ability to **bind DNA and proteins**

Challenge for their measurement :

- ✓ short half-life in the body
- ✓ high reactivity

Solution is measuring adducts

- ✓ DNA
- ✓ Proteins

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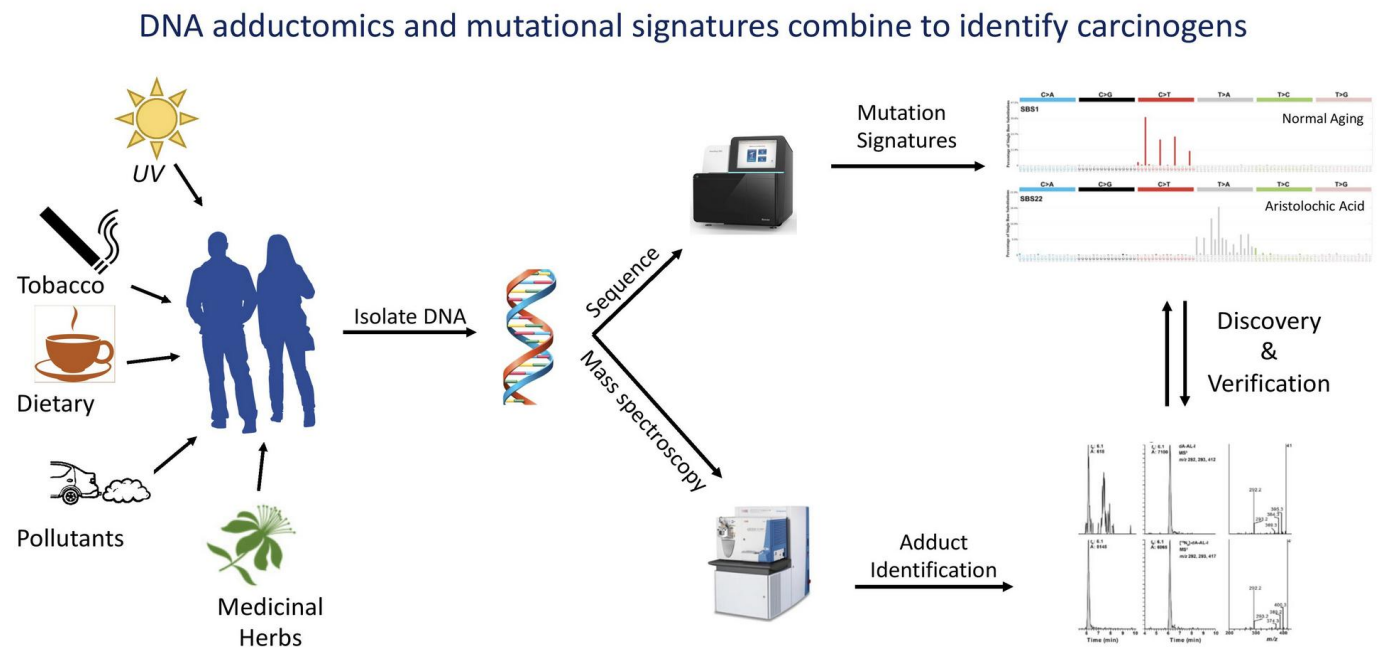
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Yun BH, Bellamri M, Rosenquist TA, Turesky RJ.
Anal Chem. Aug 21;90(16):9943-9950 (2018), PMID:30001485

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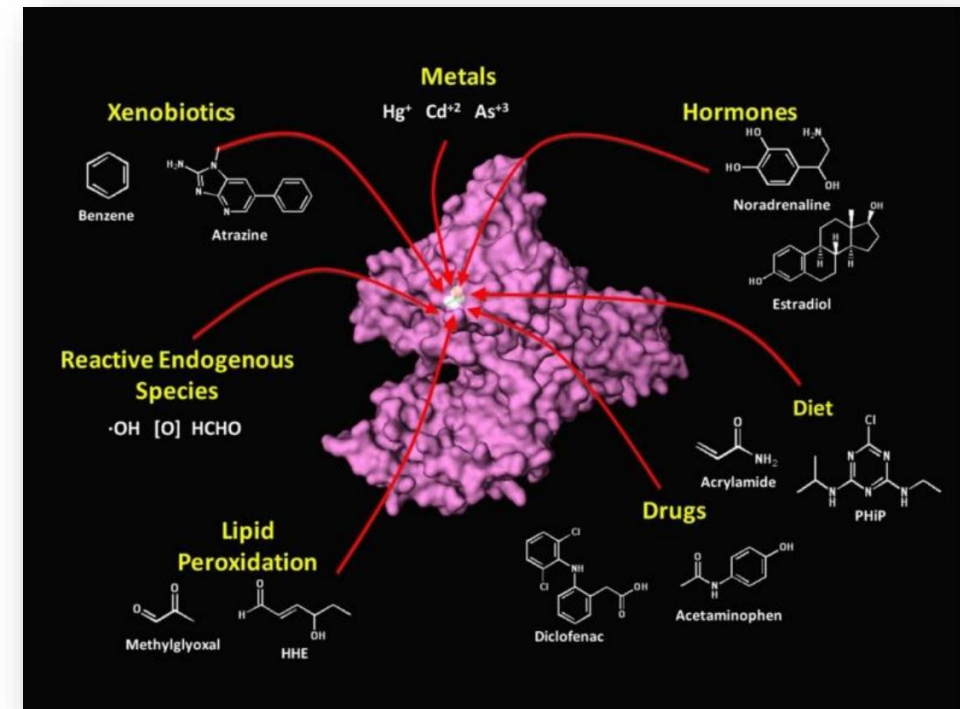
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Human Serum Albumin

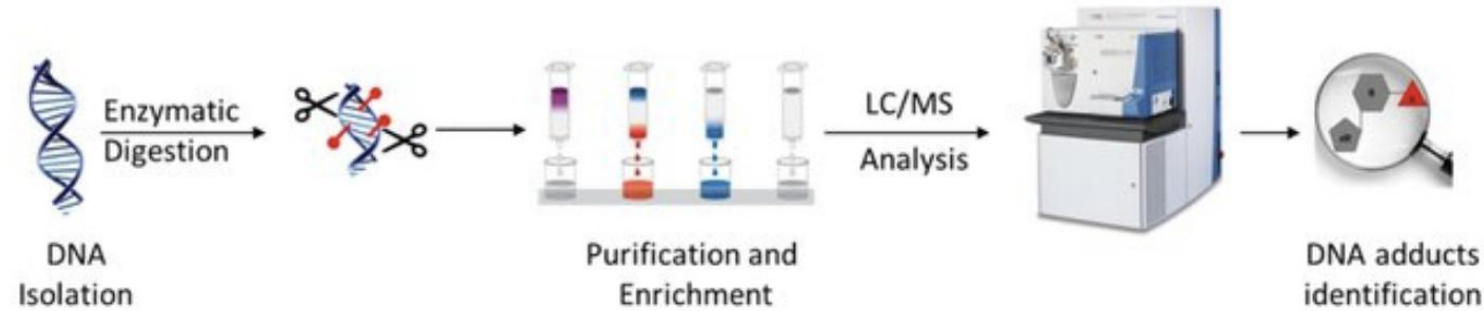
- ✓ Abundant
- ✓ Long Half-life
- ✓ Cys34 highly reactive



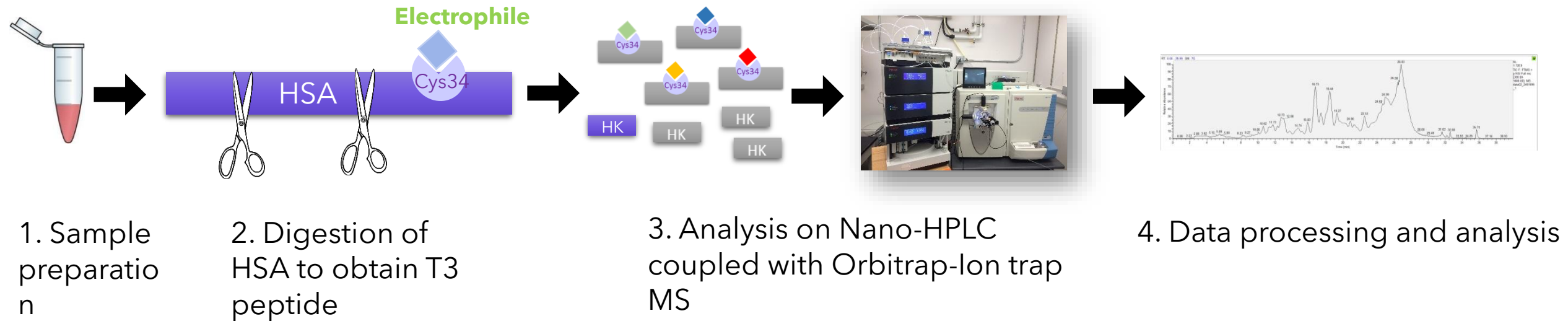
Rappaport et al. (2013) *Toxicol. Lett.* **213**, 83-90

ADDUCTOMICS: THE NON-CANONICAL ENTRY POINT

DNA



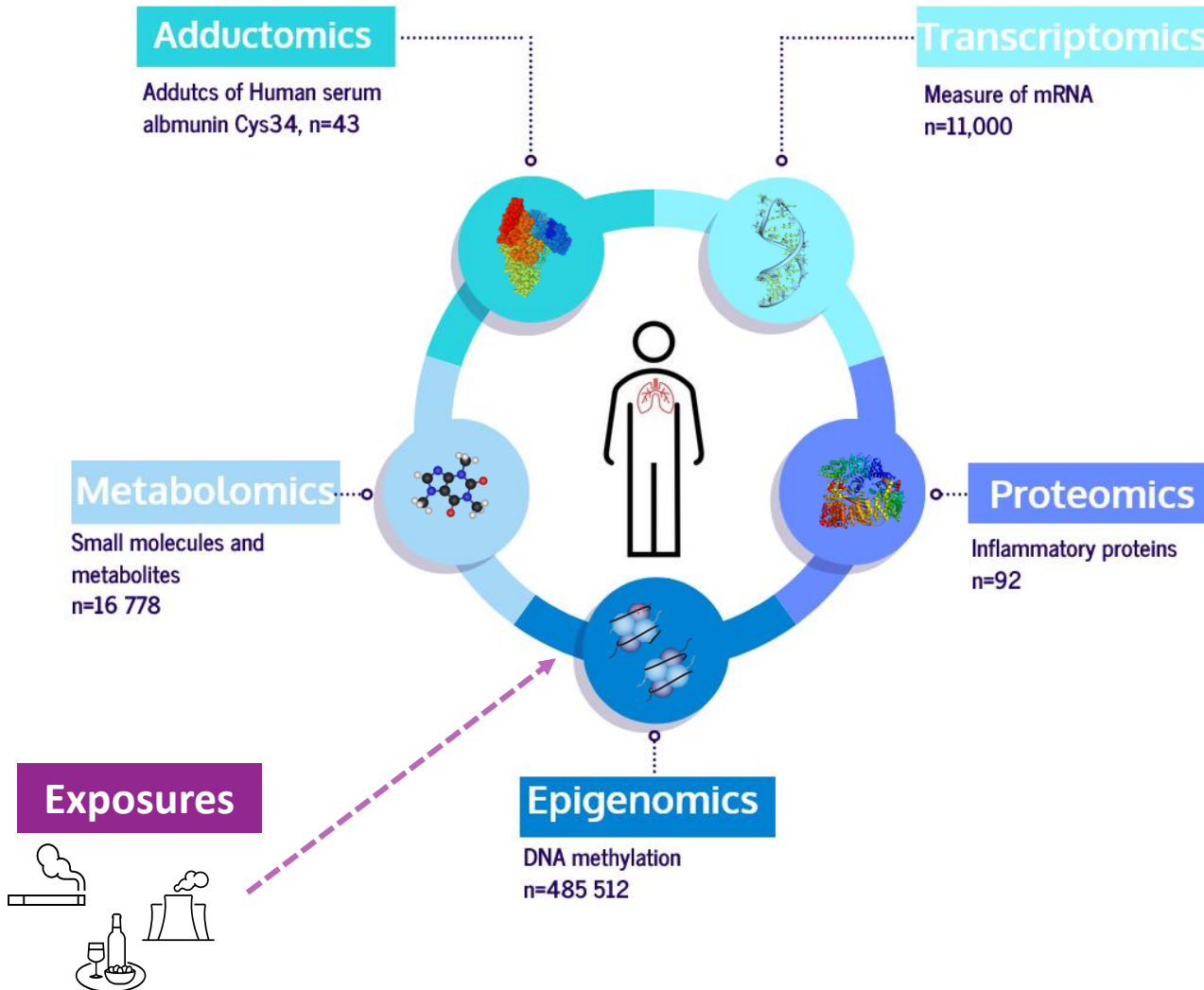
Proteins



EXPOSOME CASE STUDY



Case Study : Internal Exposome Predicts Lung Cancer



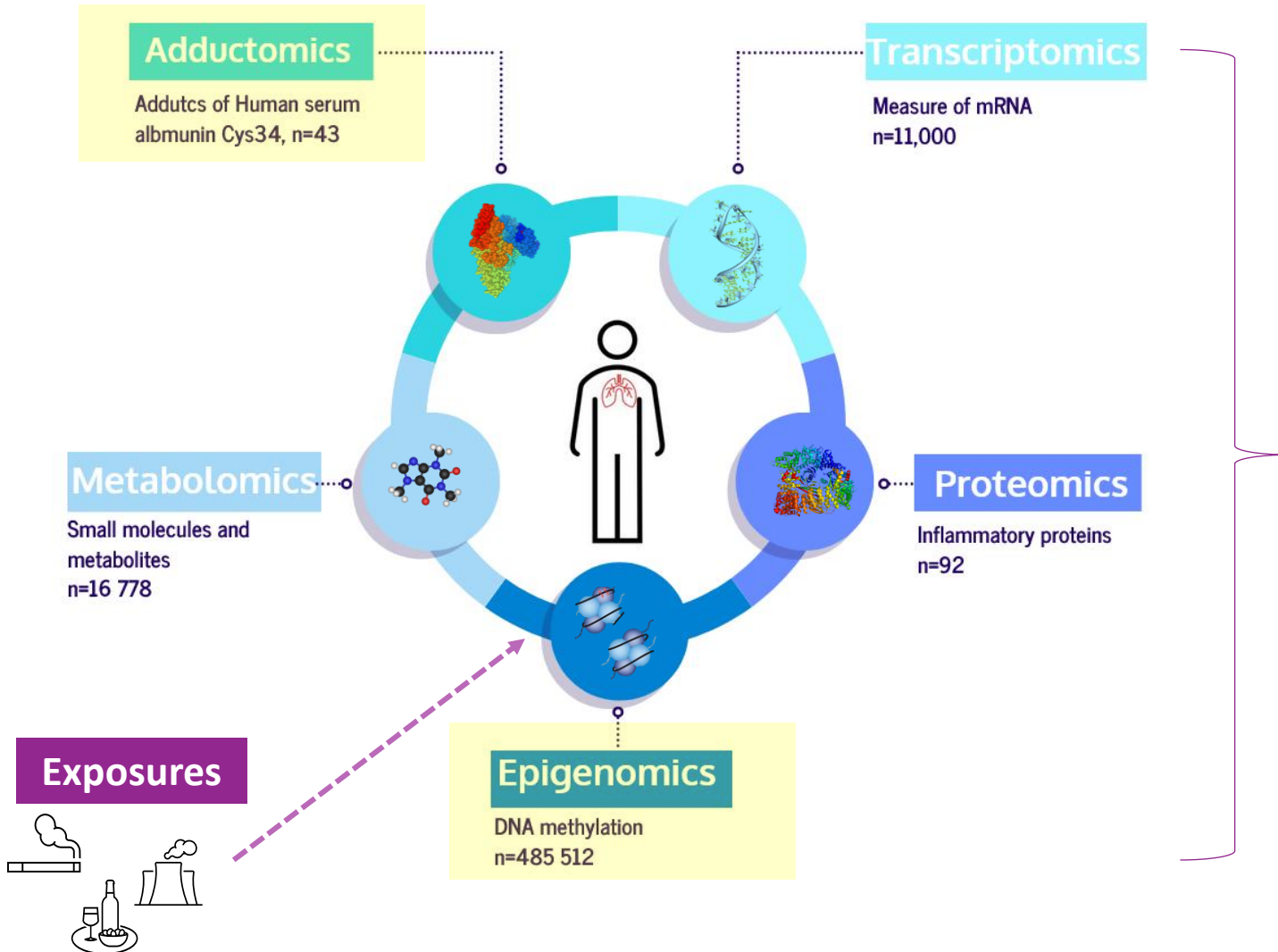
Can Internal Exposome Signatures Predict Lung Cancer

Beyond Smoking?

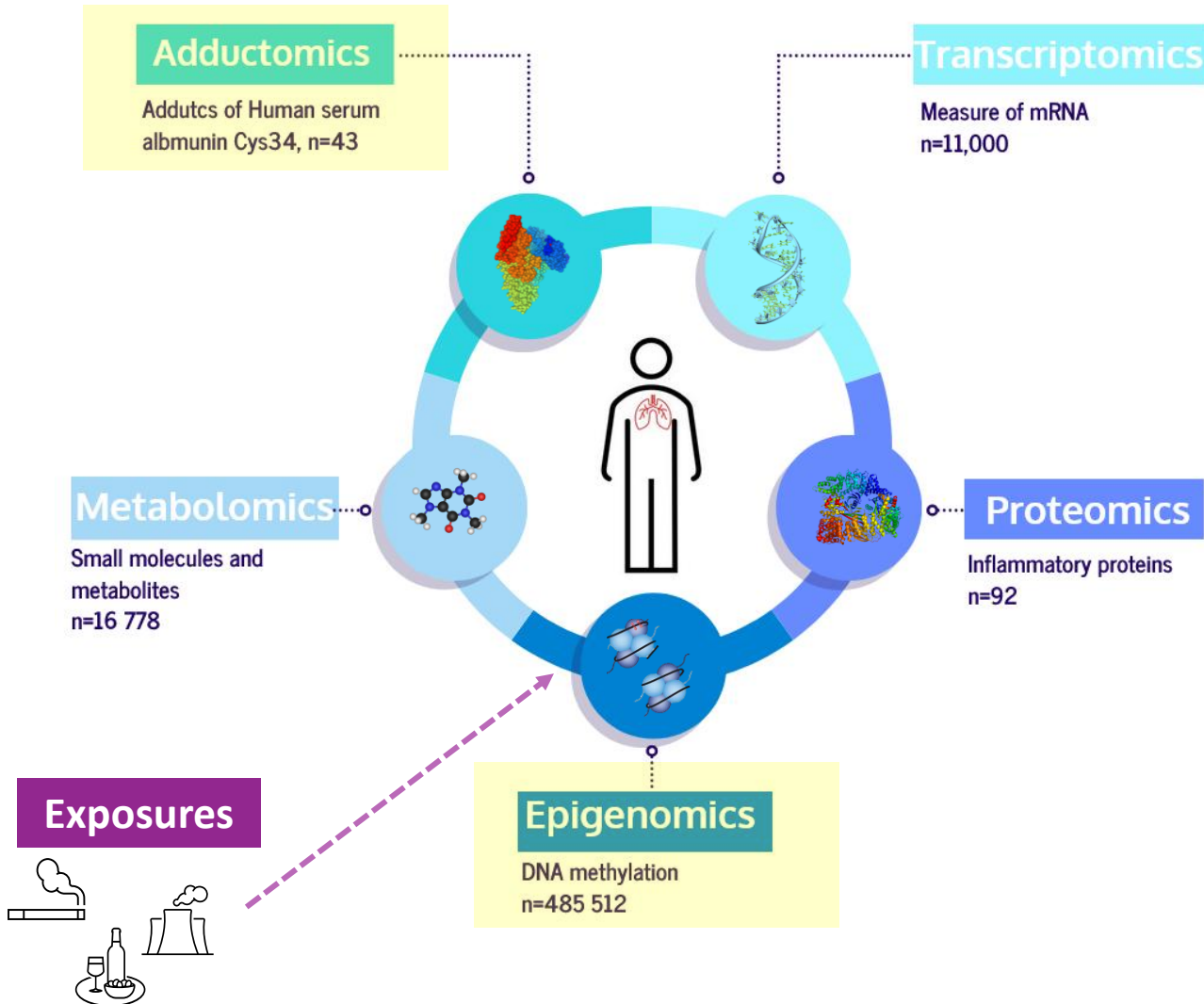
Study design:

- Prospective nested case-control
- Pre-diagnostic blood samples
- Multi-omics profiling

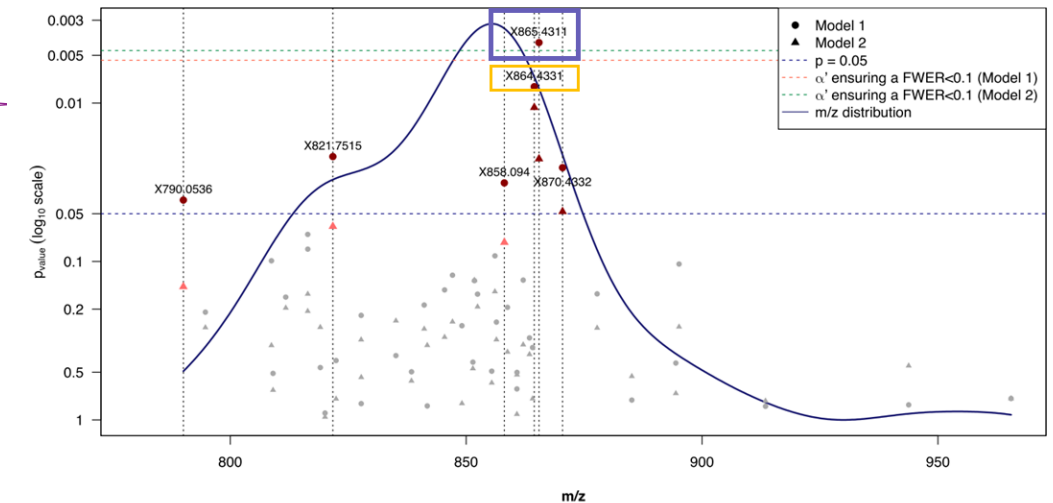
Case Study 2: Internal Exposome Predicts Lung Cancer



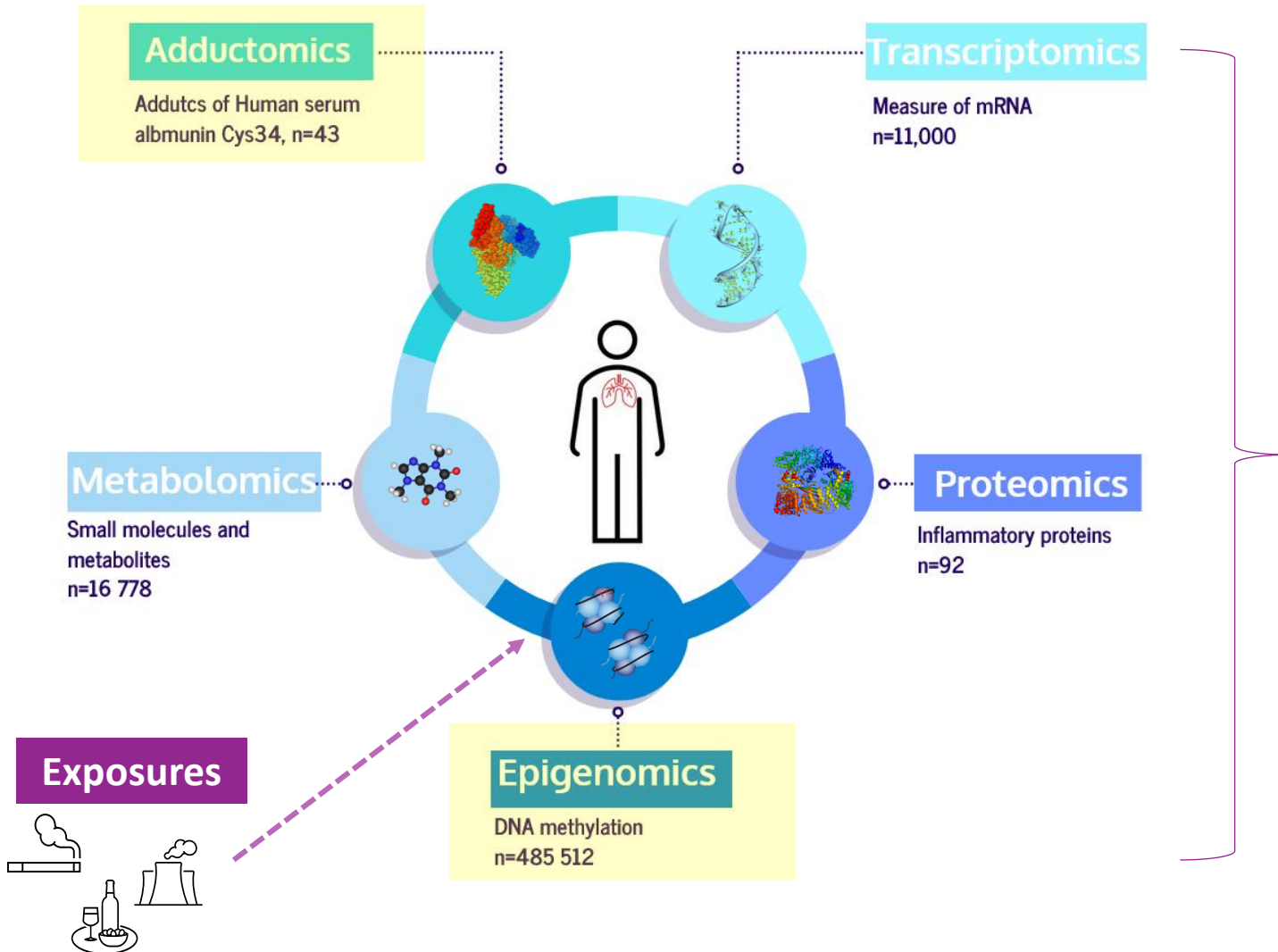
Case Study : Internal Exposome Predicts Lung Cancer



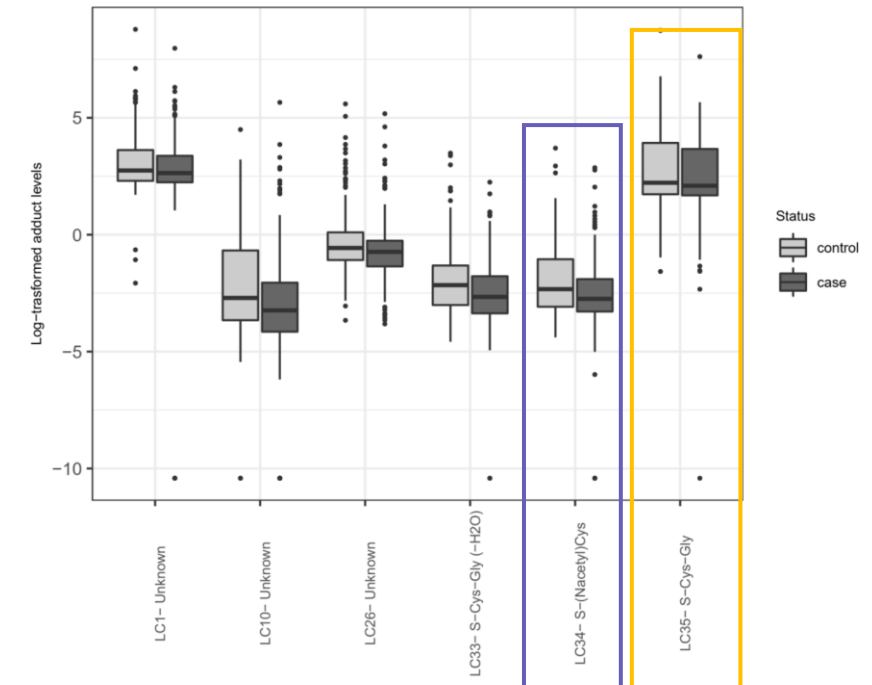
- ❑ We detected 42 unique Cys34 albumin adducts in all samples
- ❑ 6 significantly different adducts from cases and controls, years before diagnosis



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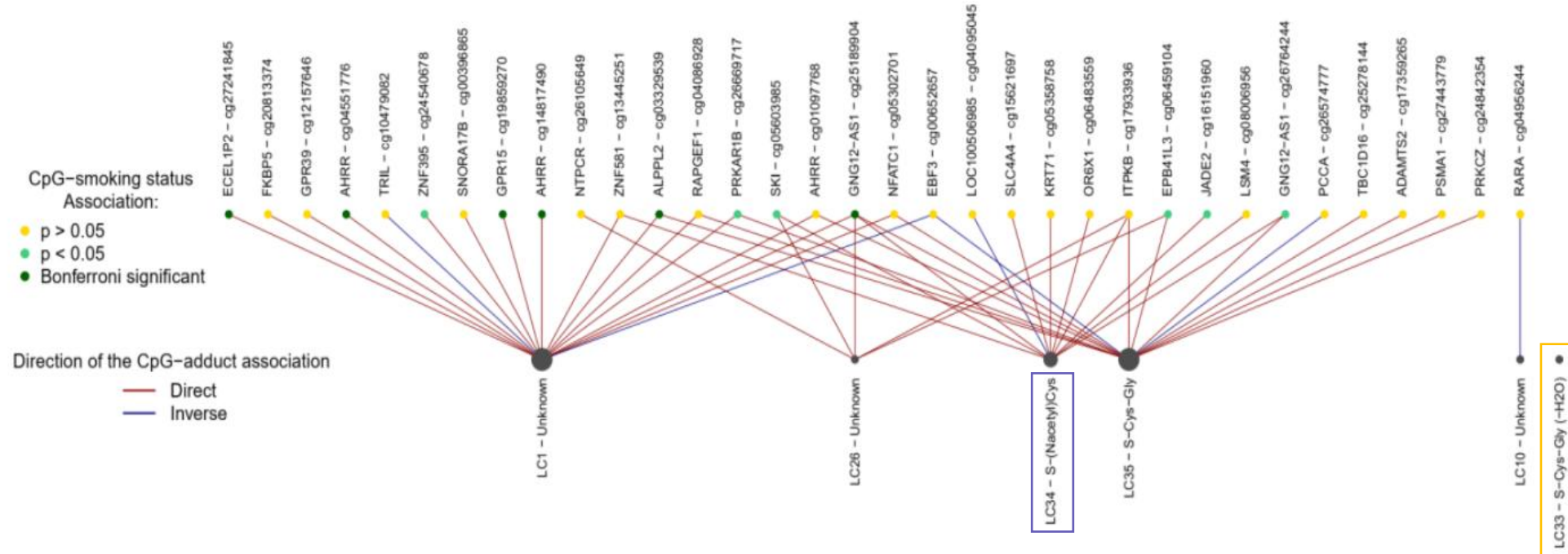


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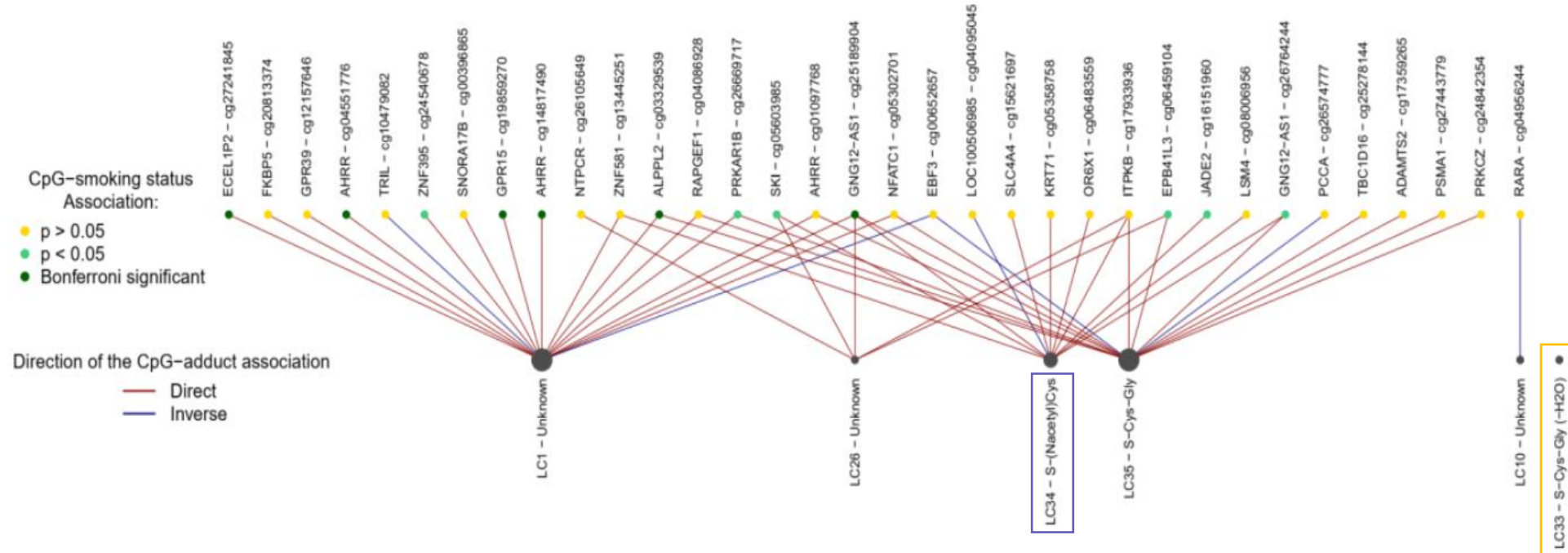
Case Study : Internal Exposome Predicts Lung Cancer

- Selected 2670 smoking related CpG sites⁹
- 11 smoking-related CpG sites associated with the NAC adduct¹⁰
- Reduced level of NAC associated to smoking-related hypomethylation
- Indicates a potential similar smoking-induced pathway



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The same exposure (smoking) leaves BOTH a protein adduct signature and an epigenetic signature. Adducts as intermediates of exposure-induced genomic alteration

CONCLUSIONS:
FROM
EXPOSURE
MEASUREMENTS
TO CAUSAL
INTEGRATION

**Canonical
Exposomics**

Exposure cataloging

Risk factor association

Biomarker discovery

**Non-
Canonical
Exposomics**

Measurement of internal reactive burden

Identification of molecular intermediates

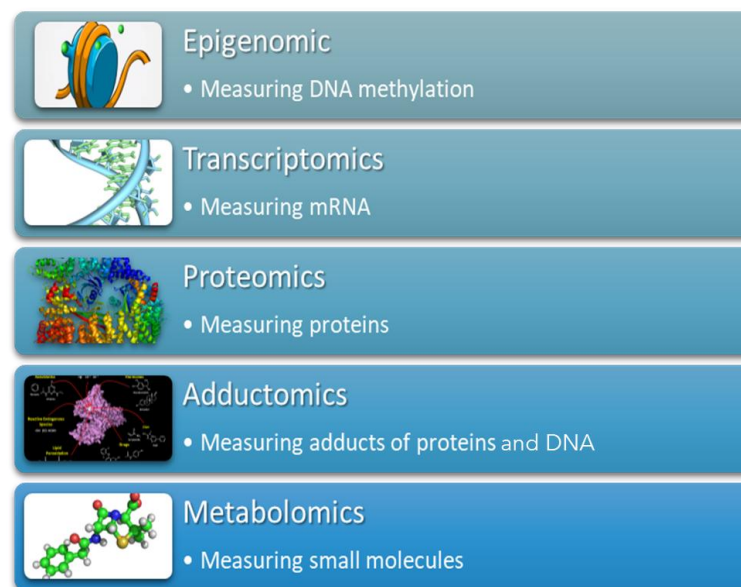
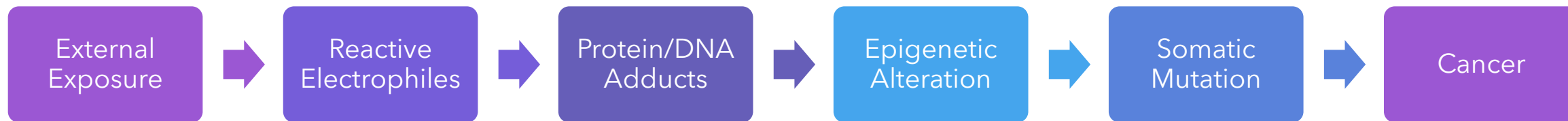
Integration with epigenetic and genomic layers

Strengthening causal inference in cancer

CONCLUSIONS:
FROM
EXPOSURE
MEASUREMENTS
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INTEGRATION

**Exposomics becomes a
mechanistic bridge
not just an exposure tool.**

EXPOSOMICS AS THE MISSING MECHANISTIC LAYER



Integrating these layers allows environmental attribution to be biologically grounded.

TAKE HOME MESSAGES



The genome tells us what mutated – not what caused it



Environmental exposures generate measurable molecular imprints



Albumin adducts and epigenetic changes can act as intermediate phenotypes



Integrating exposomics with genomics strengthens causal inference



The future of cancer epidemiology lies in cross-omics integration